

Analysis of a Doubly Spread Channel

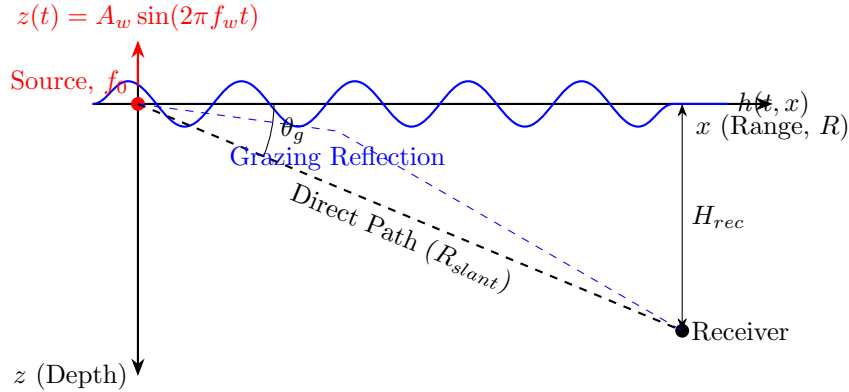
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1 Theoretical Framework

The theoretical framework defines a **Doubly Spread Channel** [1, 2] as one driven by two coupled mechanisms: **Source Kinematics (Heave)** and **Dynamic Boundary Interaction (Micro-Multipath)**.

2 Source on the surface

2.1 Schematic Diagram



2.2 Physical & Mathematical Breakdown

2.2.1 The Grazing Angle

The grazing angle θ_g is the angle between the direct acoustic path and the horizontal at the point of reflection. It is given by:

$$\theta_g = \arctan\left(\frac{H_{rec} - H_{src}}{R}\right) \quad (1)$$

Because the source is on the surface ($H_{src} \approx 0$) and the range is long ($R \gg H_{rec}$), the grazing angle θ_g approaches 0° . For example, in a setup where the receiver is at 20m depth and the source is 1km away on the surface, $\theta_g \approx 1.15^\circ$.

2.2.2 The Transmitted Signal: Source Kinematics

A theoretical underwater acoustic source emits a monochromatic pressure wave $p(t)$ at a fundamental frequency f_0 :

$$p(t) = P_0 \cos(2\pi f_0 t) \quad (2)$$

Since a surface vessel is physically coupled to the ocean surface, it undergoes vertical heave motion. We approximate the dominant ocean swell as a harmonic oscillation with amplitude A_w and frequency f_w :

$$z(t) = A_w \sin(2\pi f_w t) \quad (3)$$

This physical motion acts as a time-varying phase shift on the emitted acoustic wave. The transmitted signal $p_{tx}(t)$ becomes a Phase/Frequency Modulated (FM) signal:

$$p_{tx}(t) = P_0 \cos(2\pi f_0 t + \beta \sin(2\pi f_w t)) \quad (4)$$

The severity of the phase modulation is defined by β , which relates the physical heave amplitude to the acoustic wavelength ($\lambda = c/f_0$) and the grazing angle (θ_g):

$$\beta = \frac{2\pi f_0}{c} A_w \cos \theta_g \quad (5)$$

The instantaneous phase $\Phi(t)$ of the transmitted pressure wave is:

$$\Phi(t) = 2\pi f_0 t + \beta \sin(2\pi f_w t) \quad (6)$$

Differentiating the phase $\Phi(t)$ with respect to time yields the instantaneous frequency $f_i(t)$, defined as $f(t) = \frac{1}{2\pi} \frac{d\Phi}{dt}$. This is directly analogous to kinematics, where velocity is the time derivative of position ($v(t) = \frac{dx}{dt}$). While a constant phase rate yields a static frequency f_0 , the heave-induced sinusoidal phase results in a time-varying frequency $f_i(t)$ that physically oscillates around f_0 :

$$f_i(t) = \frac{1}{2\pi} \frac{d\Phi}{dt} = f_0 + \Delta f \cos(2\pi f_w t) \quad (7)$$

where the peak frequency deviation Δf is defined as:

$$\Delta f = \beta f_w = \frac{2\pi f_0 f_w}{c} A_w \cos \theta_g \quad (8)$$

Physical Interpretation of β : The modulation index $\beta = 2\pi(A_w/\lambda) \cos \theta_g$ represents the peak phase shift in radians, governed by two physical balances:

- **Scale Ratio** (A_w/λ): Compares the physical ship heave amplitude (A_w) to the acoustic wavelength (λ). At high frequencies (short λ), yielding a large β and severe frequency smearing. At low frequencies (long λ), resulting in a small β and minimal smearing.
- **Geometric Projection** ($\cos \theta_g$): Represents the projection of the vertical heave along the acoustic ray path. For shallow grazing angle ($\theta_g \rightarrow 0^\circ$), $\cos \theta_g \rightarrow 1$, the vertical motion projects fully along the ray path. For a receiver directly above the source ($\theta_g \rightarrow 90^\circ$), $\cos \theta_g \rightarrow 0$, the vertical motion is perpendicular to the ray direction, resulting in zero path-length modulation and no frequency smearing.

Spectral Expansion (The Bessel Comb)

[3]

To analyze the frequency spectrum of the modulated signal, we transition to the frequency domain using the **Jacobi-Anger expansion**.

First, we express the time-domain pressure wave $p_{tx}(t)$ as the real part of a complex analytic signal:

$$p_{tx}(t) = P_0 \text{Re} \left\{ e^{i2\pi f_0 t} \cdot e^{i\beta \sin(2\pi f_w t)} \right\}$$

Next, we apply the Jacobi-Anger identity, which expands the exponential of a sine function into an infinite series of Bessel-weighted harmonics:

$$e^{i\beta \sin(\phi)} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{in\phi}$$

where $\phi = 2\pi f_w t$. Substituting this expansion back into the pressure wave equation:

$$p_{tx}(t) = P_0 \text{Re} \left\{ \sum_{n=-\infty}^{\infty} J_n(\beta) e^{i2\pi(f_0 + n f_w)t} \right\}$$

Taking the real part of each term results in an infinite series of discrete spectral lines (sidebands) spaced at intervals of the ocean wave frequency f_w , weighted by Bessel functions of the first kind J_n :

$$p_{tx}(t) = P_0 \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(2\pi(f_0 + n f_w)t) \quad (9)$$

Based on **Carson's Rule**, the physical coupling of the vessel to the ocean surface theoretically expands the frequency spectrum to a bandwidth of $BW \approx 2f_w(\beta + 1)$.

2.2.3 Channel Interaction: Dynamic Boundary & Multipath

For a distant, submerged receiver, the channel impulse response $h(\tau, t)$ consists of a direct path and a surface-reflected path.

Because the source is at the surface ($H_{src} \approx 0$) and the range is long ($R \gg H_{rec}$), the geometric path length difference between the direct ray and the surface ray approaches zero ($\Delta R \rightarrow 0$), and the time delay $\Delta\tau$ between them is microscopically small ($\Delta\tau \approx 0$).

The received signal $p_{rx}(t)$ is the coherent superposition of the direct path and the surface-scattered path:

$$p_{rx}(t) = \frac{1}{R_{dir}} p_{tx}(t - \tau_{dir}) + \frac{V(t)}{R_{surf}} p_{tx}(t - \tau_{surf}) \quad (10)$$

Where $V(t)$ is the **The Time-Varying Scattering Coefficient**: [4, 5]. The reflection coefficient is not a static -1 . Instead, the moving wave height $h_s(t)$ imparts an additional time-varying phase shift $\phi_s(t)$ upon the reflected ray:

$$\phi_s(t) = 2k_{zn} h_s(t) = \frac{4\pi f_0}{c} h_s(t) \sin \theta_g \quad (11)$$

Taking the time derivative yields the Doppler frequency shift $\Delta f_d(t)$, which is directly proportional to the vertical velocity of the sea surface $v_z(t) = dh_s/dt$:

$$\Delta f_d(t) = \frac{1}{2\pi} \frac{d\phi_s}{dt} = \left(\frac{2f_0}{c} \sin \theta_g \right) v_z(t) \quad (12)$$

Thus, the scattered path is theoretically defined as:

$$V(t) = -R_0 \exp \left(i \frac{4\pi f_0}{c} h_s(t) \sin \theta_g \right) \quad (13)$$

Where R_0 is the amplitude loss due to rough-surface scattering (energy scattered out of the specular direction).

2.2.4 The Smeared Received Signal

Because $\tau_{dir} \approx \tau_{surf}$ and $R_{dir} \approx R_{surf}$, we can factor the received pressure field into a single equation that defines the phenomenon.

By substituting the FM transmitted signal $p_{tx}(t)$ and the time-varying reflection coefficient $V(t)$ into the superposition equation, the total received signal becomes:

$$p_{rx}(t) \propto p_{tx}(t - \tau_{dir}) \left[1 - R_0 \exp \left(i \frac{4\pi f_0}{c} h_s(t) \sin \theta_g \right) \right]$$

Theoretical Conclusions of the Equation: This final mathematical expression governs exactly why the signal appears smeared:

1. **The Carrier is Modulated:** The baseline signal p_{tx} is already a Bessel-expanded FM signal due to the vessel's physical heave $z(t)$.
2. **Multiplicative Fading:** The term in the brackets represents the micro-multipath interference. It acts as an **amplitude envelope** multiplying the carrier. Because $h_s(t)$ (the ocean wave height) is a random variable evolving over time, this bracketed term oscillates randomly between constructive addition and destructive cancellation.
3. **Secondary Phase Modulation:** The exponential term inside the brackets adds an additional layer of time-varying phase modulation (Doppler spread), driven by the vertical velocity of the surface waves (dh_s/dt).

The Definition of the Smear: The “smeared signal” of a surface vessel is theoretically defined as an inherently frequency-modulated tonal that is continuously subjected to multiplicative, time-varying amplitude fading and secondary phase scattering by a dynamic boundary. It cannot be mathematically reduced to a stationary sine wave.

2.2.5 Connection to the Doubly Spread Channel

Yes, this entire framework is the textbook definition of a **doubly spread acoustic channel**.

A channel is doubly spread when the transmitted signal is smeared in both **Delay** (τ) and **Doppler** (ν). The channel spreading function $\chi(\tau, \nu)$ characterizes this mathematically [1, 2].

In this specific geometry:

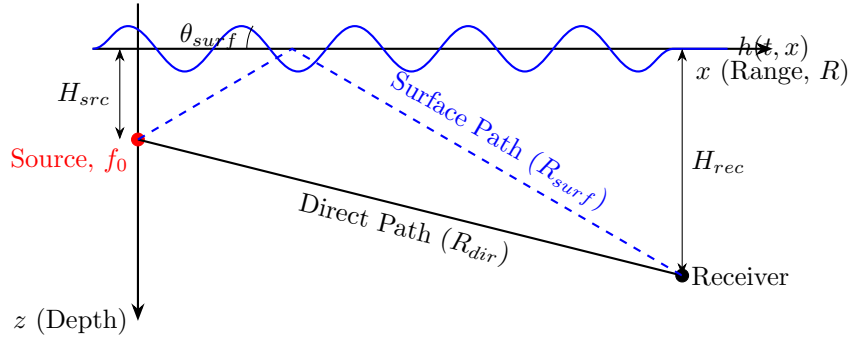
1. **Delay Spread (τ):** The extremely low θ_g forces the direct path and the surface-scattered paths to travel nearly identical distances ($R_{slant} \approx R$). The delay difference between these paths is so microscopic that they arrive within the same sampling bin, merging into a single unresolvable impulse arrival that fades constructively and destructively.
2. **Doppler Spread (ν):** Simultaneously, the dynamic velocity of the surface $v_z(t)$ and the physical heave of the source $z(t)$ convolve the signal with continuous frequency shifts, spreading the energy across the ν axis.

Because the acoustic source is physically bound to the dynamic interface, the signal is instantaneously subjected to both spreading mechanisms simultaneously before it even propagates across the range R .

3 Source Underwater

When inspecting an acoustic source deep in the water column, the acoustic physics changes because the source is decoupled from the ocean's wave spectrum, which results in a more distinct separation of geometric paths and less frequency smearing at the receiver.

3.1 Schematic



3.2 Physical & Mathematical Breakdown

When a vessel is submerged ($H_{src} \gg 0$), it is physically decoupled from the ocean's dynamic boundary. This decoupling mathematically breaks both the kinematic frequency modulation (FM) and the multiplicative micro-multipath fading.

3.2.1 Elimination of Source Kinematics

An underwater source operates below the wave base, meaning it is not subjected to the physical heave of the ocean swell.

The vertical heave amplitude approaches zero ($A_w \approx 0$). Consequently, the Modulation Index (β) collapses:

$$\beta = \frac{2\pi f_0}{c} A_w \cos \theta_g \approx 0 \quad (14)$$

Applying $\beta = 0$ to the Jacobi-Anger expansion (the Bessel comb) yields $J_0(0) = 1$ and all higher-order Bessel functions $J_n(0) = 0$. The theoretical infinite bandwidth collapses back into a single singularity. The transmitted signal $p_{tx}(t)$ reverts entirely to a pure, unmodulated monochromatic wave:

$$p_{tx}(t) = P_0 \cos(2\pi f_0 t) \quad (15)$$

meaning the source is emitting a theoretically perfect, infinitely narrow tonal.

3.2.2 Resolution of Macro-Multipath

For an underwater vessel, H_{src} is a large, non-zero value. Because of this depth, the geometry creates a significant difference in path lengths:

$$R_{surf} \gg R_{dir} \quad (16)$$

The time delay between the direct path and the surface echo becomes macroscopic ($\Delta\tau = \tau_{surf} - \tau_{dir} \gg 0$).

Because the signals no longer arrive at the same time, they can no longer interfere multiplicatively. The superposition equation is the sum of two separate, additive events in time.

3.2.3 The Decoupled Received Signal

Substituting our pure transmitted signal (p_{tx}) and our macro-delayed surface reflection into the receiver equation yields the theoretical received signal for an underwater vessel:

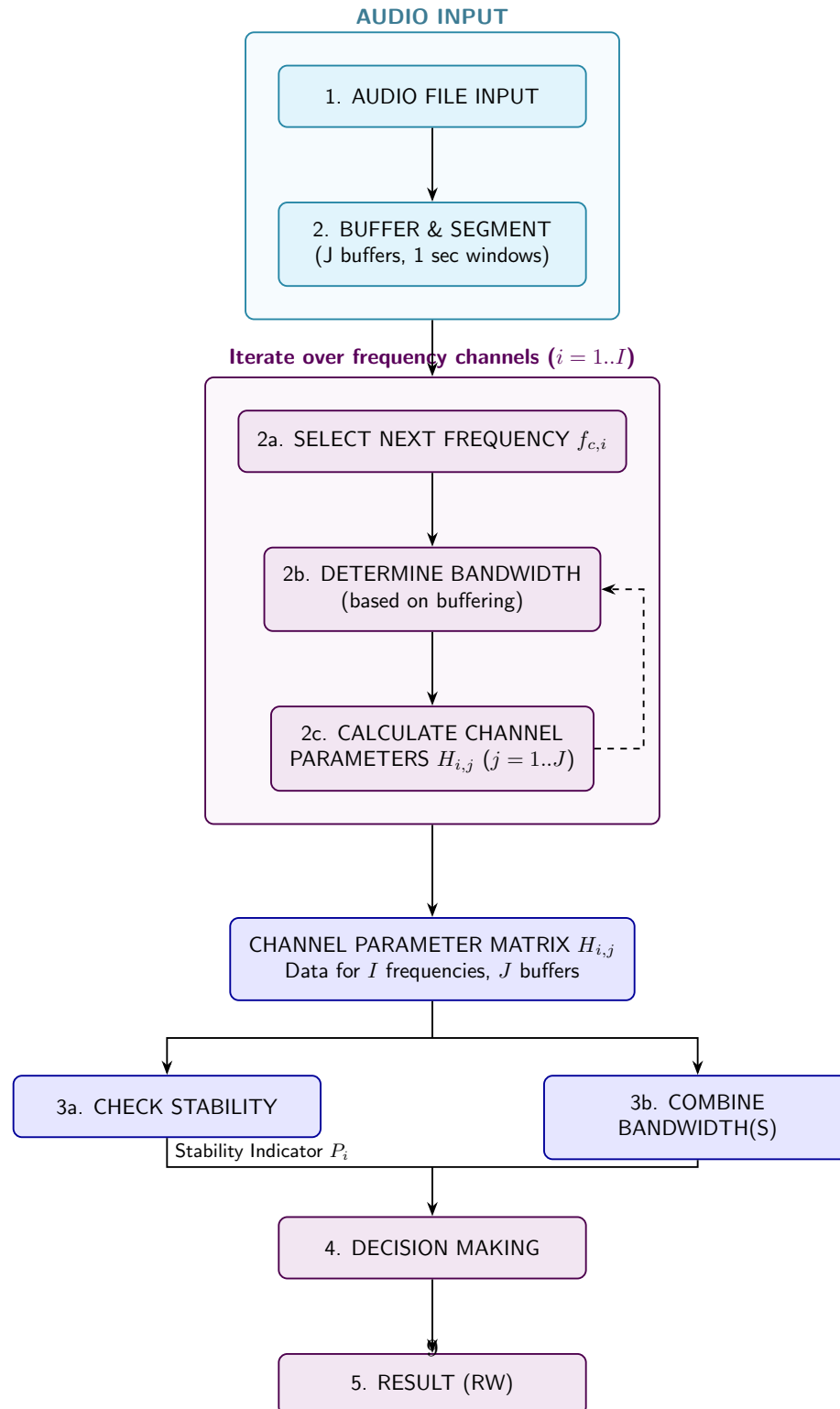
$$p_{rx}(t) = \underbrace{\frac{P_0}{R_{dir}} \cos(2\pi f_0(t - \tau_{dir}))}_{\text{Direct Path}} + \underbrace{\frac{P_0 V(t)}{R_{surf}} \cos(2\pi f_0(t - \tau_{surf}))}_{\text{Surface Echo}} \quad (17)$$

This equation perfectly defines why the main signal is not smeared:

1. **The Direct Path is Pristine:** is a perfect, time-delayed copy of the original engine tonal.
2. **The Echo is Isolated:** the second term contains the time-varying scattering coefficient $V(t)$, which means the Doppler phase modulation from the moving surface does happen to the echo. However, because τ_{surf} is significantly larger than τ_{dir} , it arrives much later.
3. **Additive vs. Multiplicative:** the echo is temporally separated, it adds background reverberation (clutter) to the environment, but it does not multiplicatively modulate or fade the primary direct-path tonal.

The geometry guarantees a pure, stable direct path that is mathematically independent of the surface dynamics. The surface smearing still exists in the environment, but it is confined entirely to a delayed, isolated echo, leaving the primary mechanical tonal sharp and un-smeared.

4 Flow Diagram



References

- [1] M. Stojanovic, “Recent advances in high-speed underwater acoustic communications,” *IEEE Journal of Oceanic Engineering*, vol. 21, no. 2, pp. 125–136, 1996.
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- [4] F. G. Bass and I. M. Fuks, *Wave Scattering from Statistically Rough Surfaces*. Pergamon Press, Oxford, 1979.
- [5] W. I. Roderick, “Doppler spectra of acoustic signals reflected from a moving rough sea surface,” Ph.D. dissertation, Yale University, New Haven, CT, 1979.